

31338 Network Servers

Lab 2b: System Updates

Aims

1. To be able to update your system

It is vital to keep your operating systems up to date with security patches and fixes. However, it is also important to ensure that your operating system is not affected by unnecessary upgrades.

Task 1: Enable Linux networking

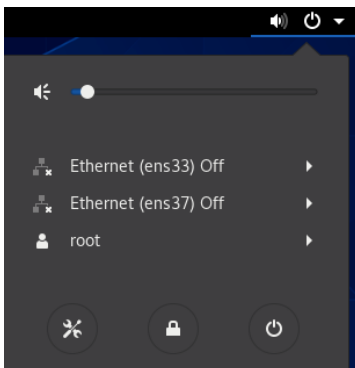
In order to install updates, your server needs to be connected to the internet (both Linux and Windows). In the Linux virtual machine supplied, networking is initially disabled (the 'NetworkManager' service). You need to do two things: start the NetworkManager service running right now and enable the NetworkManager service so that it starts every time the virtual machine boots.

You learned how to do this in Lab 1. Use the `systemctl` command to **start** the service – in `systemctl`, it is called `NetworkManager.service`

Then, run a second command using `systemctl` to **enable** the `NetworkManager.service` so that it starts up automatically when the machine boots.

Note the commands and steps in your journal (there are plenty of hints in the last 2 lines).

Once that's done, if you click on the top right of the screen you should now see a window that looks similar to below. Use the GUI to turn on Ethernet "ens33".



Verify that your network is working by ping-ing a known address (e.g. www.uts.edu.au).

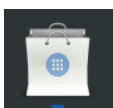
Task 2: Updates for Linux

By default, Redhat derivate Linux operating systems with a GUI use the Gnome Software manager to monitor and update system software on your Linux configuration. This is in a package called 'PackageKit'.

To start the Gnome Software Manager, run the command:

```
gnome-software
```

You can also start it from the icon that looks like a shopping bag:



Along the top of the window there should be a tab called “Updates”.
There is a “Refresh” icon that has a circular arrow. Press this to check for new updates.

Observe the list of software to update, if any. You should make a judgement on which software to install (they are ALL selected by default, but is this a good idea?)

Be EXTREMELY careful of any changes to your Kernel or GLIB – these changes can cause operational issues as you are installing.

Alternative installation method

You can use the yum command to update packages. This is often a safer way of doing updates.

Yum commands include:

<code>yum check-update</code>	(check for updates)
<code>yum search XXXXX</code>	(search for a keyword XXXXX to find packages mentioning this word)
<code>yum install XXXXX</code>	(where XXXXX is the package name)
<code>yum update XXXXX</code>	(where XXXXX is the package name)
<code>yum remove XXXXX</code>	(where XXXXX is the package name)

Yum automatically determines the dependencies and can warn you. You can also use the various options of yum to only load security or advisory fixes.

Task 3: Updates for Windows

Windows manages updates quite simply, however it is not recommended to automatically perform updates on a running server. Can you think of any reasons why??

Within Server Manager, on your “Local Server” the dashboard should show you some information about when updates were last installed and the update policy.

Last installed updates	22/07/ 10:19 AM
Windows Update	Download updates only, using Windows Update
Last checked for updates	22/07/ 11:05 AM

To run updates, you can click on these links in Server Manager, or you can go to the settings area and search for “Windows Update” in the Update & Security section.

Generally, Windows Server updates are usually well tested and can be installed. Be careful of optional updates such as .NET framework etc

You can also check for and install updates in Windows from the command line. While you can do it from the Command Prompt, Windows PowerShell provides a module for working with Windows update from the command line.

Open Windows PowerShell. To install the module, type:

Install-Module PSWindowsUpdate

It will probably prompt you / warn you about installing other dependent software – choose ‘Yes’ if asked.

Once installed, you can use PowerShell commands (“cmdlets”) like:

<code>Get-WindowsUpdate</code>	Check for updates (probably nothing if you already did it in the GUI)
<code>Get-WULastResults</code>	Show results of the last time updates were checked
<code>Get-WUHistory</code>	Show history of updates installed
<code>Get-WURebootStatus</code>	Does the system need rebooting to install most recent updates?

Try some of these and make notes in your journal. Search online for help with the “PSWindowsUpdate” module.

Run the updates now (GUI or command-line), but beware, this can take a while to download and install.